S. M. Mahbub Uddin Ahmed

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Summary

Backend and cloud engineer with research work in applied machine learning. Currently a software engineering intern at Samsung R&D Institute Bangladesh and a senior backend engineer at Fazesoft.

Experience

Samsung R&D Institute Bangladesh Dhaka, Bangladesh
Software Engineer Intern, Connectivity Part, MDE Group, IST *Mar 2026 – Present*

- Member of the Connectivity Part, the team behind the Windows version of Quick Share, the Multi Control Framework (MCF), Storage Share, Camera Share and Nearby Device.
- Built internal agentic tools and coding agents used across the engineering group, and developed Windows applications in the **.NET Framework**.
- Automated a test suite that runs across an entire device network, driving both wireless and USB-connected devices.
- **Effortless Onboarding:** an on-device AI prototype on smart glasses. Won **3rd prize at AI PoC 2026**, SRBD's internal competition between engineering groups.

Fazesoft Remote
Senior Backend Engineer *Jan 2025 – Present*

- Backend engineering on production platforms, including **HomeNet** (property marketplace, where I am system architect and lead backend engineer), **BaazarLTD**, **Passtid** and other client systems.

United International University Dhaka, Bangladesh
Undergraduate Teaching Assistant *Sep 2024 – Jan 2026, Part-Time*

- Supported faculty in core computer science and data science courses, ran lab sessions, and mentored students one-to-one.

Research & Publications

Few-shot classification of rare viruses in transmission electron microscopy images using Siamese neural networks
Preprint, SSRN 7664934 2026
A Siamese metric-learning encoder trained on the 14 common classes of the public TEM virus dataset, which recognises 8 unseen rare classes at test time without ever training on them: 94.8% accuracy on the 14-class benchmark, and 37.3% one-shot on the rare classes against a 30.8% baseline.

Is a 9B Decision Foundation Model Needed for Symptom-to-Disease Classification? Fine-Tuning Clef-Flash Against TF-IDF Baselines
arXiv:2610.15029 2026
Fine-tuned a 9B decision foundation model with 4-bit QLoRA (65.0% to 95.4%), then showed that TF-IDF linear baselines on the same split reach 98.8% at 0.7 ms per record on a laptop CPU. Includes a leave-one-symptom-out clause ablation.

BD-BANKNOTES: A Multi-Series Annotated Dataset of Bangladeshi Banknotes for Object Detection
Mendeley Data, DOI 10.17632/33vjwcrpjj.1 2026
3,466 annotated field photographs across nine denominations and two banknote series, in YOLO format with fixed 80:10:10 splits. Released under CC BY 4.0.

Education

United International University Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering; ***GPA: 3.92/4.00*** *Jan 2022 – Feb 2026*

Projects

HomeNet | *homenetbd.com*

- Property marketplace for Bangladesh with map search, verified listings and an AI listing assistant. System architect and lead backend engineer.

LiteAgent | *GitHub*

- A lightweight coding agent for small local LLMs: a deliberately minimal system prompt and tool set, because the large prompts and sprawling tool collections of current agentic systems overwhelm small models and make them hallucinate. Runs on the user's own machine, cutting reliance on paid cloud subscriptions.

CodeShare | *GitHub*

- A **.NET/C#** peer-to-peer file sharing service built for agents to drive over **MCP**: the agent discovers users on the local network by itself and sends files straight from a user prompt, with a **WPF** app for manual use.

RAGQNA | *GitHub*

- **RAG** question answering over private documents with local LLMs, keeping sensitive data on-device: vector search, retrieval and structured LLM reasoning.

Skills

Programming Languages: C, C++, C#, Java, Python, TypeScript, JavaScript, SQL.

Backend & Data: ASP.NET Core, Entity Framework Core, Node.js, FastAPI, RabbitMQ; PostgreSQL, MySQL, MongoDB, Redis, vector databases.

Cloud & DevOps: AWS, Docker, Kubernetes, Git, GitHub Actions.

AI & Research: PyTorch, TensorFlow, pandas, NLP, computer vision, metric learning, RAG, agentic systems.

Awards & Achievements

Best Code Quality Award – First Intra-University Hackathon (UIU HackDay 2026). 2026

Competitive Programming: solved almost 300 algorithm and data structure problems on LeetCode.